

CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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9231/23

October/November 2020

2 hours

You will need: List of formulae (MF19)

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

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[Turn over

- 1 (a)** By differentiating e^{-x^2} , find the Maclaurin's series for e^{-x^2} up to and including the term in x^2 . [5]

[illegible]

- (b) Deduce an approximation to $\int_0^{\frac{1}{5}} e^{-x^2} dx$, giving your answer as a rational fraction in its lowest terms. [2]

[illegible]

- 2** The variables x and y are related by the differential equation

$$9\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + y = 3x^2 + 30x.$$

- (a) Find the general solution for y in terms of x . [6]

[illegible]

- (b) State an approximate solution for large positive values of x . [1]

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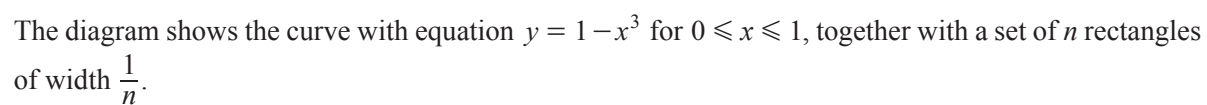
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- (c) Given instead that $k = -2$, show that the system of equations in part (a) is inconsistent. Interpret this situation geometrically. [2]

[illegible]

- (d)** For the case where $k \neq -2$ and $k \neq -4$, show that the system of equations in part **(a)** is inconsistent. Interpret this situation geometrically. [2]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.


$$\int_0^1 (1-x^3) dx \leq \frac{3n^2+2n-1}{4n^2}. \quad [4]$$
This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.

This image shows a full page of a worksheet designed for handwriting practice. It consists of multiple horizontal rows, each defined by two parallel dotted lines. The rows are evenly spaced and extend across the entire width of the page, providing a guide for letter height and placement. There is no text or other markings on the page.

(b) Find $\frac{d^2y}{dx^2}$ in terms of t , simplifying your answer.

[5]

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dotted lines, creating a series of uniform gaps for letter height. The entire page is otherwise blank, with no margins, text, or other markings.

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- (b) Find the solution of the differential equation

$$\frac{dy}{d\theta} + y \cot \theta = \sin^3 \theta$$

for which $y = 0$ when $\theta = \frac{1}{2}\pi$.

[6]

$$\mathbf{P} = \begin{pmatrix} 1 & 4 & 2 \\ 0 & -1 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

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- This image shows a full page of blank handwriting practice paper. It features multiple sets of horizontal lines, each consisting of a solid top line, a dashed middle line, and a solid bottom line. These lines are evenly spaced across the entire page, providing a guide for letter height and placement. The background is plain white, and there are no margins or additional markings.

The 3×3 matrix $\mathbf{A}$ has distinct eigenvalues $b, -1, 1$ with corresponding eigenvectors

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 4 \\ -1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix},$$

respectively.

(c) Find $\mathbf{A}$ in terms of b .

[4]

[illegible]

$$\coth^2 x - \operatorname{cosech}^2 x = 1. \quad [3]$$
[illegible]

The curve C has equation $y = \ln \coth\left(\frac{1}{2}x\right)$ for $x > 0$.

- (c) Show that $\frac{dy}{dx} = -\operatorname{cosech} x$. [3]

[illegible]

- (d) It is given that the arc length of C from $x = a$ to $x = 2a$ is $\ln 4$, where a is a positive constant.

Show that $\cosh a = 2$ and find, in logarithmic form, the exact value of a . [7]

[illegible]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting or typing. There are no margins, text, or other markings on the page.

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